Probabilistic CT Liver Atlas for Distance to Vasculature Estimation

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ABSTRACT

Vasculature infiltration is suspected to be the final stage of cancer, accompanied by the swift deterioration of major organ systems over a short period of time. Once a tumor reaches the vasculature, the prognosis changes dramatically, making the distance between a tumor and the surrounding vascular structures a potentially critical clinical indicator. This vasculature infiltration is the primary focus and index of this project, in which we seek a method to approximate the distance to the vasculature from a single patient's CT scan. Rather than registering one patient's anatomy directly onto another's, all subjects are aligned into a shared coordinate frame and accumulated into a density model, analogous to an electron cloud, that captures where liver tissue and vasculature typically exist across a population. For the purposes of this project, we are ignoring angiogenesis, and the project's function is to identify major veins rather than afferent capillary structures. Additionally, this project is currently focused only on the liver, but there is room to expand to other organs in the future. As such, the project focuses on the distance to the vasculature, specifically the hepatic and portal veins.

The pipeline for this project uses TotalSegmentator to automatically extract the liver parenchyma, all eight Couinaud segments, the portal vein, and hepatic vein segments from standard CT volumes retrieved from the TotalSegmentator website. Volumes are rigidly aligned using centroid-based translation and NIfTI affine direction normalization, and accumulated into sex-stratified probabilistic atlases for male and female cohorts. Rigid alignment improved median Dice similarity from 0.146 to 0.724 across 261 subjects. Distance to vasculature maps were computed per subject in native space and warped into atlas coordinates, yielding population-level mean distance estimates for every voxel in the atlas space. The resulting atlas provides a foundational framework for estimating tumor to vasculature proximity in new patients.

CCS CONCEPTS

• Applied computing; Health informatics; Computing methodologies; Image registration; Shape analysis; Image segmentation

KEYWORDS

Medical Image Registration, Automated Segmentation, Liver Atlas, Vascular Anatomy Analysis, Anatomical Correspondence, Health Informatics

1 INTRODUCTION

1.1 Clinical Motivation

Vascular infiltration represents one of the most consequential events in solid tumor progression. In pancreatic cancer, the tendency of tumors to invade major blood vessels is a primary determinant of surgical resectability and patient prognosis [1]. Recent work has demonstrated that invading tumor structures closely track vessels and nerves during tissue infiltration, rather than directly converging on them, revealing coordinated spatial invasion patterns [1]. This observation suggests that spatial proximity to vascular structures may represent an intermediate stage in the invasion process. Consequently, estimating the distance between a tumor and surrounding vasculature may provide a useful quantitative proxy for assessing progression toward vascular infiltration, although this relationship remains to be fully established.

This project focuses specifically on the liver, targeting the hepatic and portal venous systems as the primary vascular structures of interest. Angiogenesis and afferent capillary structures are explicitly out of scope; the focus is on the major veins whose infiltration would represent a critical threshold in hepatic cancer progression.

1.2 Problem Statement and Scope

Existing approaches to tumor-to-vessel distance estimation typically operate on a pairwise basis. Essentially, the objective here is to register one patient’s CT scan directly

onto another one or onto a fixed reference patient. In practice, however, this is not reliably achievable for smooth, largely featureless organs such as the liver. This project addresses the problem differently, by constructing a population-level probabilistic atlas that captures where liver tissue and vasculature exist across a cohort of subjects, avoiding the need for pairwise anatomical correspondence entirely.

This project is additionally entirely organ-specific, focusing on the liver, with the hepatic and portal venous systems as the primary vascular targets. However, extension to other organs is discussed as future work in section 7.2.

1.3 Contributions

This work makes the following contributions:

- An end-to-end automated pipeline for liver atlas construction from standard clinical CT volumes, using TotalSegmentator for segmentation and rigid alignment into a canonical coordinate frame
- Probabilistic atlases encoding population-level tissue density across male and female cohorts
- Population-level distance to vasculature maps for both the hepatic and portal venous systems, computed in native space and warped into the atlas coordinates

2 Literature Review

2.1 Medical Image Registration and Similarity Transforms

Medical image registration encompasses a broad family of methods for aligning two or more images into a common coordinate frame [2]. These methods are commonly distinguished by the degrees of freedom of their underlying deformation models, which range from low-dimensional rigid transformations to high-dimensional nonparametric deformations[2]. At the most constrained end, rigid registration allows only translation and rotation, preserving the anatomy's shape and scale entirely. Affine registration includes both scaling and shearing, thus allowing asymmetrical stretching while preserving parallel lines. At the most flexible end, non-rigid or deformable registration allows voxels to move independently, enabling one shape to be fully deformed to fit another. This is the class of methods that includes Thin Plate Spline interpolation [3] and iterative closest point approaches. Non-rigid methods offer the greatest anatomical flexibility but require reliable point correspondence between the source and target. This is a fundamental limitation when working with smooth, largely featureless three-dimensional objects such as an organ.

This project implements rigid registration through centroid translation combined with a canonical rotation derived from each subject’s NIfTI affine direction matrix. This choice deliberately preserves the overall shape and volume of the liver across the population, since the atlas is designed to study where tissue and vasculature exist across subjects rather than to warp one subject's anatomy onto another's. In addition, a similarity transform, which would add an isotropic scaling factor to the rigid alignment, was evaluated; however, it was ultimately omitted to preserve natural size variation as a meaningful anatomical signal.

2.2 Probabilistic Atlases and Population-Level Analysis

A probabilistic atlas is a population-level anatomical model that encodes the frequency with which a structure appears at each location in a shared coordinate space, rather than treating any single subject as a definitive anatomical reference [4]. Early work in probabilistic atlases focused primarily on the brain, where population-level maps were used to capture normal structural variability and to support automated segmentation and lesion localization, using a database of over 7000 patients. The core idea here was to align many subjects into a common frame and accumulate their anatomy into a density representation. This idea has since been extended to other organs in the abdomen, where inter-subject variability in position, size, and orientation is a consideration [5,6]

In this project, the probabilistic atlas functions as a spatial probability density over liver tissue: each voxel in the shared grid stores the fraction of the population for whom liver tissue was present at that location, producing a continuous map analogous to an electron cloud. High-density regions represent locations where liver tissue is consistent across subjects, whereas lower-density regions reflect natural anatomical variability. This approach avoids the template bias introduced by single-subject reference models and preserves inter-subject variation as a meaningful population signal. This idea is then extended by attaching population-level vascular distance statistics to each consensus liver location, thereby providing a generalizable basis for estimating tumor-to-vasculature proximity in new patient scans.

2.3 Liver Segmentation and Landmark-Based Registration

TotalSegmentator [7,8] is a publicly available nnU-Net-based toolkit trained on over 1228 clinically diverse CT examinations, providing automated segmentation of 104 anatomical structures, including the liver parenchyma, hepatic and portal vasculature, and the Couinaud liver segments[9,10]. Its broad anatomical coverage and robustness across scanner types and contrast phases made it practical for segmentation in this project. However, TotalSegmentator occasionally produces incomplete or empty masks for smaller structures, necessitating the quality control steps described in section 3.1.

Landmark-based registration approaches for the liver have frequently relied on the Couinaud segmentation scheme as an anatomical reference framework, using segment boundaries, portal vein bifurcations, and hepatic vein branches as control points for establishing inter-subject correspondence[11]. In the original design of this pipeline, Couinaud segment centroids were used as landmarks for TPS-based alignment. As described in section 3.4, this approach was abandoned due to the fundamental correspondence problem, and the current pipeline derives its alignment schema from the NIfTI affine direction matrix for canonical orientation, with centroid-based translation providing coarse positional alignment. Couinaud segment boundaries remain a promising avenue for refining intra-atlas correspondence and a promising topic for future discussion in section 7.2.

3 Methodology

3.1 Dataset and Preprocessing

CT volumes were sourced from the publicly available TotalSegmentator dataset [7], which comprises over 1228 clinically acquired scans representing a diverse patient population across multiple sites and scanners. Because the dataset reflects real-world clinical imaging rather than a controlled study protocol, a manual integrity verification step was performed prior to any processing. A scripted review pipeline was developed to render and display the 3D liver surface for each patient, enabling visual inspection of each liver. Cases that exhibited artifacts such as truncation, where the liver volume was partially cut off at the scan boundary, were entirely excluded from the dataset. Following this initial screening process, 303 cases were retained (122 female, 179 male, and 2 with unknown sex labels).

Anatomical segmentations required for downstream atlas registration were not included in the original dataset release and were therefore generated using the TotalSegmentator toolkits [7,8,9,10]. Following segmentation, a second automated quality control pass was applied to detect any silent failures, cases where the TotalSegmentator worked but failed to produce a segmentation. In such cases, TotalSegmentator could have produced an empty or nearly empty output masks for either the Hepatic veins, Portal veins, or Couinaud liver segmentations. In cases where empty masks were created, these patients were additionally removed from the pool. After both filtering steps, 263 subjects were usable. Two cases had unknown sex labels and were excluded from sex-stratified atlas construction, yielding an analytic cohort of 261 subjects (107 female, 154 male).

From here, the volumes were padded into a common array grid based on the maximum observed volume extents to prevent anatomical cropping during downstream alignment and accumulation.

3.2 Automated Segmentation via TotalSegmentator

Anatomical segmentations were generated using three separate TotalSegmentator subtask calls, each invoking a different underlying model because the main TotalSegmentator model doesn't support all of the different types of segmentations we used. The first call used the main TotalSegmentator model with the –roi_subset flag to extract only the liver parenchyma and the combined portal/splenic vein mask; this subset restriction significantly reduces runtime on CPU hardware. The second call invoked the liver_segments tasks [10] to produce the eight Couinaud segment masks. The third call invoked the liver_vessels task, whose output was used as the hepatic vessel mask for downstream distance mapping [9]. All outputs were saved per patient for downstream use.

It should be noted that while the liver_vessels task also produces a liver tumor mask, the TotalSegmentator dataset is not a dedicated tumor cohort, and tumor masks were not used in the atlas pipeline. Subjects identified as having hepatic lesions were flagged and excluded from the primary cohort but retained for future comparative work. The portal and splenic vein segmentation from the first subtask and the hepatic vein segmentation from the third subtask serve as the primary vasculature targets for the distance mapping described in Section 3.6.

3.3 Rigid Alignment in a Canonical Frame

Alignment was performed in two passes, using one manually selected artifact-free reference subject per sex cohort to establish an initial common frame. Pass 1 cached per-subject centroids, extents, and affine-direction metadata. Between passes, these records were used to derive several population-level normalization terms: a global bounding box encompassing all warped patient extents, a global coordinate offset ensuring any negative coordinates mapped to valid array indices in the shared output grid, a median liver voxel count, and a cohort-level canonical orientation matrix derived from the stored per-patient affine directions.

This information was used in the second pass, in which each patient was realigned using the population-level terms. The rotation component is derived from the canonical NIfTI affine direction matrix, and translation was applied using the whole-liver centroid as the anchor point. All warped liver masks were then accumulated into a single shared global grid, forming the basis for the probabilistic atlas described in section 3.5. Isotropic scaling, while a standard component of Procrustes similarity alignment[12], was intentionally omitted to preserve natural size variation across the population, as liver volume differences represent meaningful anatomical variation among subjects.

3.4 Approaches Considered and Abandoned

Initial research into organ-to-organ registration mapping pointed toward established methods in non-rigid registration. Methods such as Iterative Closest Point (ICP) and Thin Plate Spline (TPS) interpolation seemed like cohesive solutions to a seemingly impossible problem. TPS was explored as a candidate approach because it can theoretically deform the geometry of one subject's liver onto another's, which seemed promising for addressing the distance-to-vasculature problem.

The fundamental limitation of this approach, however, was the correspondence between specific points. TPS requires that each point on the source map be uniquely correspond to a point on the target. In practice, no reliable method exists for establishing true anatomical correspondence between arbitrary points on two different livers. The organ has some features, and one could identify certain key points as landmarks; however, 100% precise correspondence was impossible. There are simply not enough landmarks to reliably establish a method for a direct correspondence between two separate organs from separate subjects and to meaningfully align them.

In response to this limitation, the pipeline was entirely redesigned around what is effectively a population density model, an approach that is conceptually similar to that of an electron cloud, where, rather than trying to non-rigidly deform one liver onto another one, all the livers were accumulated into a shared grid to produce a probability map of where liver tissue exists across the population. This atlas captures population-level anatomical tendency without requiring pairwise point correspondence. Precise tumor location within this new atlas space, which would require a solution to the correspondence problem, potentially using Couinaud segment boundaries as anatomical anchors for triangulation, remains an open problem and is discussed further in section 7.2 as a direction for future work.

3.5 Atlas Construction

The atlas was constructed across two separate passes as described previously in section 3.3. The outputs of Pass 1 were per-patient alignment cache files and a JSON record of each patient's spatial extents and affine direction, which were used between passes to compute the shared global grid. In Pass 2, each patient's aligned liver mask was warped into this grid and accumulated into a density volume. The density volume is normalized by the total patient count; each voxel value represents the fraction of the population for whom liver tissue was present at that location, on a scale from 0 (no patients) to 1 (all patients). A consensus liver boundary was defined by thresholding this density volume, retaining voxels where tissue was present in a majority of all subjects. This consensus surface aims to represent the population-level liver shape and serves as the reference geometry for the distance-to-vasculature mapping described in Section 3.6. In this case, the threshold was set to 0.5 for the current implementation, meaning any voxel present in at least half of the population is included in the consensus boundary. It is, however, highly desirable to tighten this threshold to one to two standard deviations above the mean density, as it would produce a more conservative consensus surface and is noted as a refinement for future work.

3.6 Distance-to-Vasculature Cloud

Distance-to-vasculature mapping was first computed in each subject's native space, then warped into atlas space. For each liver voxel, the mean Euclidean distance to the five nearest vessel voxels was computed separately for each vessel system (portal/splenic and hepatic) using a k-nearest neighbors approach (k=5) implemented as a k-D tree[13]. From there, the mean distance to the five nearest neighbors was calculated rather than to the single nearest neighbor to reduce sensitivity and error, thereby preventing randomly isolated, disconnected voxels from skewing the calculation. Three per-patient maps were produced: one for the portal/splenic vein, one for the hepatic vein, and a combined map taking the minimum distance to either vessel system per voxel, which was then averaged across the population during accumulation.

The resulting native-space distance maps were then warped into the shared atlas grid using the rigid alignment parameters cached during the atlas build. Because distance maps are continuous scalar fields, trilinear interpolation was applied at this step during the warp in order to preserve smooth transitions between neighboring voxel values. This is in contrast to the binary liver masks warped during atlas construction, which used nearest-neighbor interpolation to keep the mask values strictly binary. Warped distance maps were accumulated across all patients and averaged at each atlas voxel, producing a population-level distance-to-vasculature map sampled over the consensus liver region defined in section 3.5. Hence, every liver voxel in the consensus liver has a mean population distance to both the portal and hepatic vasculature, yielding complementary population-level vascular distance representations (portal/splenic, hepatic, and nearest-system combined) within a shared atlas coordinate system.

4 Evaluation Framework

4.1 Dice Similarity Coefficient

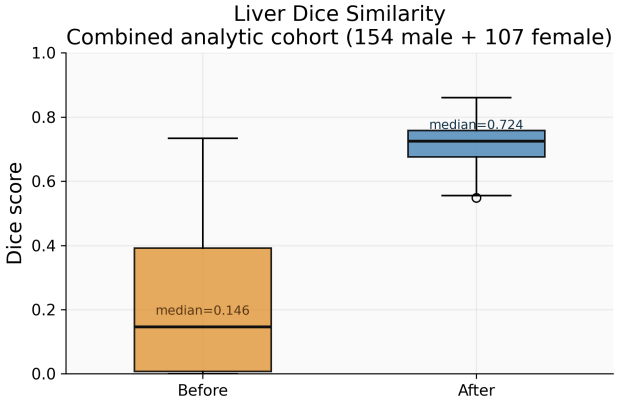

Figure 1. Liver Dice similarity coefficient before and after rigid alignment across the combined analytic cohort (154 male + 107 female subjects, n=261). Pre-alignment median DSC = 0.146; post-alignment median DSC = 0.724.

Registration quality was evaluated using the Dice similarity coefficient (DSC), which was computed between each source patient's liver mask and the reference subject's liver mask before and after alignment, as shown in Figure 1. A DSC of 0 indicates no overlap, and a DSC of 1 would be indicative of a perfect overlap. Prealignment Dice scores had a mean of 0.216 (median 0.146; range 0.000–0.735; n=261), reflecting the large positional and orientational variation across subjects in native scanner space. Post-alignment Dice scores improved substantially, with a mean of 0.717 (median 0.724; range 0.548–0.861), indicating substantial improvement in volumetric overlap following the introduction of rigid alignment routines.

4.2 Centroid Spread Statistics

Prior to alignment, centroid-anchor displacement between the source and reference subjects averaged 296.98 mm (median 131.18 mm, SD 314.03 mm; range 18.20–1579.60 mm; n=261), reflecting the substantial positional variability across the clinical dataset. Post-alignment centroid-anchor distance was 0 mm by construction of the rigid transform, confirming that the translational component of alignment was applied correctly across all subjects.

4.3 Rotation Angle Verification

Following canonical affine-direction alignment, effective rotation angles were computed across 261 subjects. Rotations were generally small, with a median of 0.097 degrees and a mean of 0.603 degrees (SD 1.223 degrees; range 0.037–7.152 degrees; 95th percentile 3.469 degrees), confirming that the NIfTI affine direction matrix was successful in normalizing subject orientations, with any residual need for rotations remaining sub-degree for the majority of the cohort.

4.4 Qualitative Visualization

Alignment quality was further assessed through qualitative visualization, including overlay views of aligned liver masks, 3D surface renderings, and before-and-after vessel-alignment panels. Average liver shell anatomy checks confirmed the presence of the expected porta hepatis indentation on the consensus surface, providing anatomical plausibility for the population atlas.

5 Results

5.1 Registration Quality

Rigid alignment substantially improved volumetric overlap across the combined analytic cohort of 261 subjects (154 male, 107 female), as shown in Figure 1. Prealignment Dice scores had a median of 0.146 (mean 0.216; range 0.000–0.735), reflecting large inter-subject positional variability in native scanner space. Post-alignment Dice scores improved significantly, with a median of 0.724 (mean 0.717; range 0.548–0.861), confirming that rigid centroid-based alignment successfully brought liver volumes into a common coordinate frame.

5.2 Population Atlas

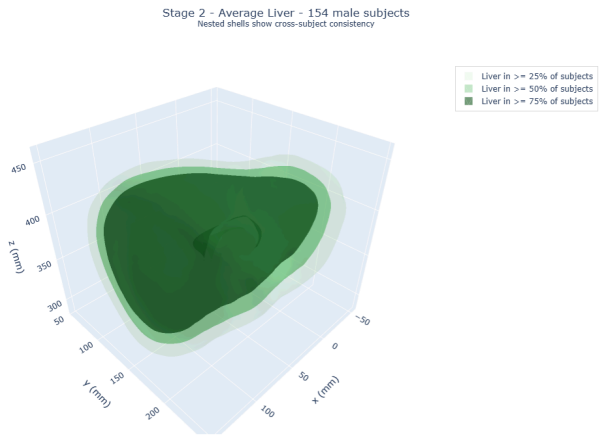

Figure 2. Population-level probabilistic liver atlas for the male cohort (n=154 subjects), displayed as nested density shells at the 25%, 50%, and 75% population thresholds. The porta hepatis indentation is visible in the high-density inner shell.

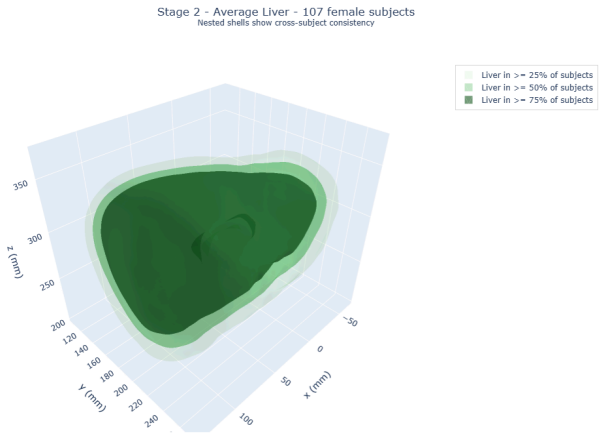

Figure 3. Population-level probabilistic liver atlas for the female cohort (n=107 subjects), displayed as nested density shells at the 25%, 50%, and 75% population thresholds.

Probabilistic atlases were constructed separately for male (n=154) and female (n=107) cohorts. Figures 2 and 3 show the nested density shells for each cohort at the 25%, 50%, and 75% population thresholds. Both atlases preserve the expected gross morphology of the liver, including the characteristic porta hepatis indentation visible in the inner high-density shell. This provided anatomical plausibility validation of the alignment and accumulation pipeline.

5.3 Distance-to-Vasculature Maps

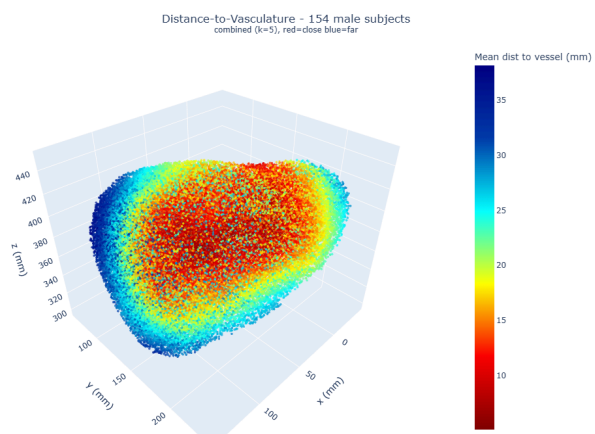

Figure 4. Population-level combined distance-to-vasculature map for the male cohort (n=154 subjects). Each consensus liver voxel is colored by mean distance to the nearest vessel system (portal/splenic or hepatic), with red indicating proximity (~5–10 mm) and blue indicating greater distance (~39 mm).

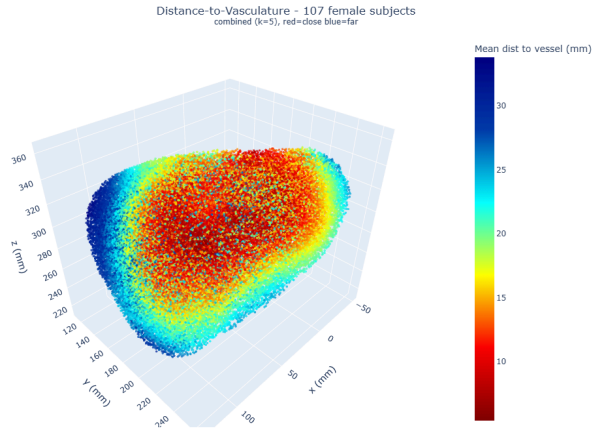

Figure 5. Population-level combined distance-to-vasculature map for the female cohort (n=107 subjects). Color scale same as Figure 4; maximum distances are slightly lower than the male cohort (~34 mm).

Population-level distance-to-vasculature maps were computed for both cohorts, as shown in Figures 4 and 5. The combined distance map, which represents the mean distance to the nearest vessel system across the population, shows a very consistent pattern in both of the sexes. The voxels that are closest to the central region of the liver, where the portal and hepatic vasculature converge at the porta hepatis, showcase the smallest mean distances (~5-10mm), while peripheral liver voxels show the greatest distances (up to about 39mm in males and 34mm in females). The distance distribution across consensus liver voxels followed a right-skewed pattern with a mean of 18.29mm and a median of 17.5mm for the male cohort (n=467,894 voxels, range 5.06-38.90mm). As for the female cohort, it also followed a similar distribution, with a mean of 16.48mm and a median of 15.80mm (n=389,109 voxels, range 5.14–33.81 mm), as shown in Figures 6 and 7.

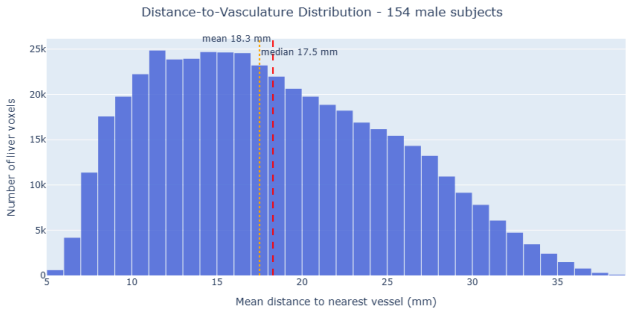

Figure 6. Distribution of mean distance-to-vasculature across consensus liver voxels for the male cohort (n=467,894 voxels; mean 18.29 mm, median 17.50 mm; range 5.06–38.90 mm).

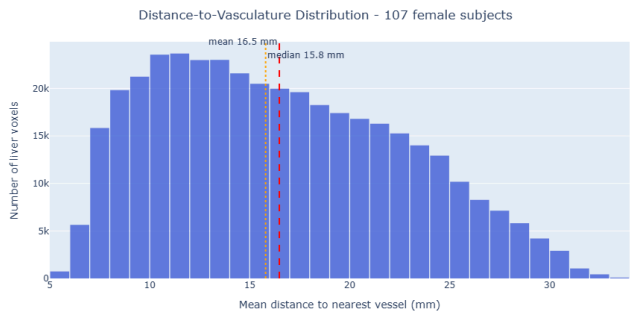

Figure 7. Distribution of mean distance-to-vasculature across consensus liver voxels for the female cohort (n=389,109 voxels; mean 16.48 mm, median 15.80 mm; range 5.14–33.81 mm).

6 Discussion

6.1 Interpretation of Registration Quality

The improvement in the Dice similarity coefficient from a pre-alignment median of 0.146 to 0.724 post-alignment demonstrates that rigid centroid-based alignment substantially reduced inter-subject positional variability across the cohort. For context, the state-of-the-art segmentation model on which this project is based, TotalSegmentator, achieves a Dice score of approximately 0.94 [7]. These values, however, are computed by comparing predicted segmentation to ground truth within the same image, rather than across subjects. In contrast, registration-based Dice scores reflect alignment between anatomically different individuals and are therefore inherently lower and more difficult to interpret directly. This kind of Dice score between two different subjects' CT scans would only be achievable with a model that uses non-rigid deformation to ensure that the voxels in one model align with those in the other. This was, however, intentionally excluded from this pipeline in order to preserve the natural anatomical variation across the subject population.

It should be noted that Dice measures volumetric overlap rather than direct anatomical correspondence; essentially, two liver volumes can overlap substantially without their internal structures (such as the vasculature) aligning precisely[2]. For the purposes of this atlas, which aims to capture population-level tissue density and vascular distance distributions rather than establish pairwise anatomical correspondence, the achieved alignment is considered more than sufficient. For context, atlas-based segmentation methods using cascaded rigid and deformable registration for anatomically complex structures have reported Dice scores around 0.789 [14], suggesting that the median 0.724 achieved with rigid-only alignment in this pipeline is competitive, given the additional constraint of preserving natural size variation across the population.

6.2 Interpretation of Distance-to-Vasculature Results

The spatial pattern observed in both the male and female distance maps is anatomically consistent with expectations. Shorter mean distances in the central liver region reflect the higher density of portal and hepatic venous structures converging at the porta hepatis, while longer distances at the peripheral margins reflect the relative scarcity of major vessels in those regions. This gradient from the core to the periphery is what one might expect, given the liver's vascular architecture. The presence of this gradient in the population-level atlas provides qualitative support for the pipeline's anatomical validity.

The observed difference in mean distances between the male (18.29mm) and female (16.48mm) cohorts likely reflects overall liver volume rather than any fundamental difference in vascular architecture. The male and female consensus livers differ by approximately 100,000 voxels at 1.5mm spacing, representing a substantial volumetric difference. A smaller liver will, on average, place any given parenchymal voxel closer to the nearest vessel simply due to the reduced organ size.

6.3 Limitations

The most significant limitation of the current pipeline is the lack of independent clinical validation for the individual segmentation masks used to compute distances to the vasculature model. The underlying segmentation models used to extract hepatic and portal vascular structures have been validated in the original TotalSegmentator publication and subsequent follow-up studies [7,9]. The accuracy of those segmentations within this specific cohort and their downstream effects on the distance maps have not been reviewed by a radiologist. The anatomical plausibility of the atlas model is supported by the presence of expected structures, such as the porta hepatis indentation. However, formal validation against expert-annotated vasculature masks would certainly be necessary before these distance estimates could be used in any real clinical context.

The cohort used to build the atlas is also worth acknowledging. Subjects were drawn from the TotalSegmentator dataset, which reflects a specific clinical population from hospitals in the Basel, Switzerland system. The dataset was drawn from routine clinical CT examinations acquired across multiple years at the University Hospital Basel [7], making it a clinically relevant but geographically specific population. In addition to this selection, this project filtered to exclude truncation artifacts, surgical resection, and tumors. The resulting atlas, therefore, represents a relatively healthy liver population. Factors such as cirrhosis, fatty liver disease, or prior surgical intervention can meaningfully alter the liver's size, shape, and vascular distribution, and the current atlas

should not be generalized to pathological populations without further validation.

It is worth noting that the distance maps capture only major hepatic and portal venous structures as segmented by TotalSegmentator. Smaller vessels, including venules and capillaries, may not be resolved at this scale and are not included in the distance computation. This is a relatively meaningful caveat, as the capillary and venule substructures may actually be more vulnerable to vascular infiltration due to their overall permeability and proximity to tumor margins. Incorporating finer vascular resolution would require either higher-resolution imaging or dedicated small-vessel segmentation models, both beyond the scope of the current pipeline. Additionally, it may be beneficial to include MRI data, as MRI images better showcase boundary conditions and may more accurately locate the capillaries. This is an item for further discussion in section 7.2.

6.4 Relationship to Dr. Ghersi's Broader Project

Clinically, the primary motivation for this distance mapping is the hypothesis, central to Dr. Ghersi's broader project [1], that distance to vasculature may predict time to vasculature infiltration in tumors. A tumor located 5mm from any major vessel represents a fundamentally different clinical situation than one located 30mm away. A tumor close to the vasculature may infiltrate it sooner and enable metastatic spread significantly earlier than one farther from the vasculature. The population-level distance atlas developed here provides a reference framework for contextualizing such measurements across a patient population.

This pipeline is best understood as a foundational stepping stone rather than a complete clinical tool. The project originated from a core question posed by Dr. Ghersi: Can we determine the distance to vasculature across CT scans for any specific patient? The atlas attempts to answer that question at the population level. What remains is the patient-facing side. Essentially, what needs to be developed now is a method to map a new patient's tumor location into this atlas space and read off the corresponding population-level distance estimate at that location. This would require solving the point correspondence problem discussed in Section 3.4, likely through the implementation of Couinaud segment-based triangulation, and is the primary direction for future development described in Section 7.2. Until that patient query framework is implemented and the vasculature masks are validated by a radiologist, this pipeline serves as a prototype demonstrating the feasibility of the population-level approach.

7 CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project presents an end-to-end automated pipeline for constructing a probabilistic liver atlas and a population-level distance-to-vascular map from clinical CT volumes. Beginning with over 1228 publicly available CT scans, the pipeline applies automated segmentation via TotalSegmentator, rigid alignment to a canonical coordinate frame, and probabilistic accumulation of liver tissue and vascular distance measurements across the cohort. The result from this is two liver atlas models, one for male subjects and one for female subjects, each encoding both the population-level distribution of liver tissue and the mean distance to the nearest hepatic or portal vessel at every consensus liver location.

In the process of this work, there was a dramatic shift from pairwise CT to CT registration toward a population-level probabilistic model. Rather than directly deforming one patient's anatomy onto another, the atlas model aggregates information across subjects into a shared coordinate space, producing a generalizable reference map of vascular proximity. This atlas model will hopefully serve as a foundational component of Dr. Ghersi's broader clinical project, which hypothesizes that distance to vasculature may be a good predictor of time to vasculature infiltration in tumors. The pipeline demonstrates that population-level distance mapping is feasible from standard clinical CT data and establishes the infrastructure on which a patient-facing tumor distance-to-vascular framework can be built.

7.2 Future Work

The most immediate priority is implementing the patient query module, which maps a new patient's tumor location into atlas space and retrieves the corresponding population-level distance estimate. This requires a solution to the point correspondence problem, most likely through the usage of triangulation and landmarks. Using Couinaud segments to create an atlas model that preserves this level of delineation, along with markers for other specific landmarks on the liver, would allow new patients to be reliably mapped into the atlas coordinate frame. Effectively enabling the ability to calculate the approximated distance to vasculature for any new patient.

Independent radiologist review of the vasculature segmentation masks and resulting distance estimates is also a necessary step before this tool can be used for any clinical applications. While TotalSegmentator's segmentation models have been validated in prior work, their accuracy within this specific cohort and the downstream fidelity of the distance maps have not been formally assessed by a clinical expert.

The integration of MRI data represents another promising direction. MRI provides superior soft-tissue contrast and boundary definition compared with a CT scan, potentially enabling more accurate localization of smaller vascular substructures, such as venules and capillaries, that are not reliably resolved in standard CT volumes. A multimodal

pipeline incorporating both CT and MRI could substantially improve the fidelity of the distance maps.

A further refinement to the distance mapping pipeline would involve filtering the vasculature masks to remove substructures smaller than the diameter threshold for true veins. Major veins such as the hepatic and portal vein may typically range from 5-20mm in diameter, while venules measure approximately 10-20 micrometers and capillaries 5-10 micrometers. Applying a Sobel filter to detect vessel boundaries and computing an approximate diameter per connected component would allow the pipeline to dynamically include or exclude certain structures. This would produce a much cleaner and clinically meaningful distance map. Additionally, it may actually alter the distance gradient between the central vascular core and the peripheral liver parenchyma.

Finally, while the current pipeline is focused exclusively on the liver, the underlying atlas construction approach is organ-agnostic and could certainly be extended to other abdominal organs, such as the pancreas and kidneys, so long as adequate segmentations are available. However, the problem of tumor proximity is localized and should be treated as such.

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AI Usage Disclosure:

For this research paper, I used Grammarly to ensure my writing and grammar were clear. Additionally, I used Claude to create the references, help outline the paper, and to ensure that my writing was clear and getting the point across effectively/quickly.